LIFTING A MAP FROM A SUMMAND OF A FINITE FREE MODULE ALONG A SURJECTION, LOCALLY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: the reduction to the free case, the translation into lifted sections and the local liftability are all inside the counted span.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology for a ringed space. That statement is the special case of this ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The single labelled reference occurring inside the counted spans points to a chapter of the Project not staged in this card's source tree; it is carried in assembly.external_references as a non-load-bearing navigation pointer recorded at the invoking line.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Direct summand of a finite free module on a ringed site. Same chapter, the notion of a direct summand of a finite free O_U-module, which is the concrete form of a finite projective module. Supplies the hypothesis on E, and the reduction to the case E = O_U^{direct sum n}.
- Characterisation of surjective maps of sheaves. Sites, the characterisation of an epimorphism of sheaves by local liftability of sections. Supplies the step that the lift of the images of the basis elements is possible on the members of a covering.
- Sections of a finite free module and their description by basis elements. Same chapter, the identification of Hom(O_U^{direct sum n}, F) with tuples of sections of F over U. Supplies the equivalence between a dotted arrow and a lift of the images of the basis elements.

Source reference. Original section 00WL: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 11792–11792. Cross-chapter navigation pointer invoked at proof lines 11791-11792, where the author uses the characterisation of surjective maps of sheaves. The target chapter is not staged in this card's source tree, so the pointer records the invoking line; it only resolves the reference for the reader, and no bibliography citation is involved.

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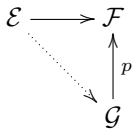
Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let U be an object of $\mathcal{C}$. Given a solid diagram of $\mathcal{O}_U$-modules*



with $\mathcal{E}$ a direct summand of a finite free $\mathcal{O}_U$-module and p surjective, then there exists a covering $\{U_i \to U\}$ such that a dotted arrow making the diagram commute exists over each U_i.

Proof. We may assume $\mathcal{E} = \mathcal{O}_U^{\oplus n}$ for some n. In this case finding the dotted arrow is equivalent to lifting the images of the basis elements in $\Gamma(U, \mathcal{F})$. This is locally possible by the characterization of surjective maps of sheaves (Sites, Section 00WL). □